

# Joongheon Kim

Last update on April 27, 2026

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## Educational Backgrounds

- **University of Southern California (USC) – Viterbi School of Engineering** *Los Angeles, California, USA*
    - Ph.D. (08/2009–08/2014) in **Computer Science** (Advisor: Prof. Andreas F. Molisch, Department of Electrical Engineering) Communications, Information, Learning & Quantum (CLIQ) Group
    - M.S. (05/2014) in **Computer Science** with specialization in **High Performance Computing and Simulations**
    - M.S. (05/2012) in **Electrical Engineering**
  - **Korea University – College of Informatics** *Seoul, Republic of Korea*
    - M.S. (03/2004–02/2006) in **Computer Science and Engineering** (Advisor: Prof. Wonjun Lee)
    - B.S. (03/1999–02/2004) in **Computer Science and Engineering**
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## Professional Affiliations

### Full-Time

- **Korea University**, Seoul, Republic of Korea
  - Professor (03/2026–Present)**, School of Electrical Engineering (Graduate: Department of Electrical and Computer Engineering)
  - Associate Professor (03/2021–02/2026)**, School of Electrical Engineering
  - Assistant Professor (09/2019–02/2021)**, School of Electrical Engineering
  - Adjunct Professor (03/2023–Present)**, Department of Communications Engineering (Samsung Electronics)
  - Adjunct Professor (03/2021–Present)**, Department of Semiconductor Engineering (SK Hynix)
  - Adjunct Professor (11/2022–Present)**, Department of Future Science and Technology Business (Graduate School)
  - Affiliated Professor (03/2025–Present)**, Department of Defense Convergence Technology (LIG Nex1, Graduate School)
- **Chung-Ang University**, Seoul, Republic of Korea
  - Assistant Professor (03/2016–08/2019)**, School of Computer Science and Engineering
- **Intel Corporation**, Santa Clara (Silicon Valley), California, USA
  - Systems Engineer (09/2013–02/2016)**, Platform Engineering Group
- **University of Southern California – Viterbi School of Engineering**, Los Angeles, California, USA
  - Ph.D. Student in Computer Science (08/2009–08/2014)**, Thomas Lord Department of Computer Science
  - Annenberg Graduate Fellow (2009)**, 4-Year Full Scholarship for the Ph.D. Program in Computer Science
  - Research Assistant (01/2011–08/2014)**, Communications, Information, Learning & Quantum (CLIQ) Group
  - Teaching Assistant (01/2012–05/2013)**, Computer Science and Electrical Engineering Major Courses
- **InterDigital**, San Diego, California, USA
  - Intern (05/2012–08/2012)**, Wireless Systems Evolution Department
- **LG Electronics – CTO Office**, Seoul, Republic of Korea
  - Research Engineer (01/2006–08/2009)**, Multimedia Research Lab, Seocho R&D Campus

### Joint Appointments for Quantum Research Collaboration

- **Seoul National University Hospital**, Seoul, Republic of Korea
  - Visiting Professor (10/2025–Present)**, Healthcare AI Research Institute

### Services for National R&D

- **Presidential Council on National Artificial Intelligence Strategy**, Advisory Committee (12/2025–Present)
  - **National Research Foundation of Korea**, Review Board (11/2024–10/2026)
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## University R&D/Administrative Positions

### R&D Leadership Positions

- **Korea University**, Seoul, Republic of Korea
  - Director (07/2024–12/2031)**, Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center
  - Principal Investigator (07/2024–12/2031)**, Software Star-Lab (Quantum AI Empowered Second-Life Platform Technology)

### Administrative Positions

- **Korea University**, Seoul, Republic of Korea
  - Vice Department Chair (01/2025–08/2025)**, Academic Affairs, School of Electrical Engineering
  - Deputy Vice President (02/2022–08/2024)**, Office of Academic Affairs
  - Dean (06/2021–08/2023)**, Center for Teaching and Learning (CTL)

### Committees

- **Korea University**, Seoul, Republic of Korea  
**Member (12/2023–Present)**, Council of the Interdisciplinary Program in Quantum Information Science, Department of Physics  
**Member (03/2021–02/2027)**, Korea University 120th Anniversary Planning Committee  
**Member (06/2021–03/2022, 09/2022–08/2023)**, Academic Affair Planning Committee  
**Member (03/2021–08/2023)**, Distance Learning Management Committee  
**Steering Committee Member (07/2020–06/2022)**, Institute of Data Science (IDS)

## Awards and Honors

### Research and Academic Excellence (International)

- **Top Cited Article in ETRI Journal (Wiley), among work published in 2024** 2025
- **Finalist, AAAI Student Abstract and Poster Session – Oral Presentation** 2026 (Top 35), 2023 (Top 25)
- **IEEE ICTC Best Paper Award – IEEE Communications Society** 2022
- **Spotlight, Oral Presentation – ICML Workshop on Dynamic Neural Networks (2022)** 2022
- **IEEE MMTC Best Journal Paper Award – IEEE Communications Society** 2021
- **IEEE ICOIN Best Paper Award – IEEE Computer Society** 2021
- **IEEE MMTC Outstanding Young Researcher Award – IEEE Communications Society** 2020
- **IEEE Systems Journal Best Paper Award – IEEE Systems Council** 2020
- **Next Generation and Standards (NGS) Division Recognition Award – Intel Corporation** Q1/2015
- **Annenberg Graduate Fellowship Award – University of Southern California** 2009–2013  
Awarded with Ph.D. Admission: 4 Year Full Scholarship (Tuition Waiver and \$120,000 Stipend (\$30,000/year for 4 years))
- **IEEE Seoul Section Student Paper Contest** Gold (2019), Silver (2025), Bronze (2025, 2024, 2024, 2023, 2020)
- **IEEE VTS Seoul Chapter Award – IEEE Vehicular Technology Society** 2023, 2022, 2022, 2021, 2021, 2019

### Research and Academic Excellence (Korea Regional, since 2016)

- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2025
- **Best Paper Award, The Journal of KICS – KICS** 2024
- **HFR Paper Award (Area: Quantum Technologies and Quantum Communications) – KICS** 2023
- **Haedong Young Scholar Award – KICS and Haedong Foundation** 2018  
For recognizing a researcher under the age of 40 who has made outstanding contributions to IT R&D

### Korea University

- **Granite Tower Best Research Award (Top 3%) – Korea University** 2024
- **Best Research Achievement Award – Korea University, School of Electrical Engineering** 2024
- **Insung Research Grant Award – Korea University** 01/2023  
For recognizing professors in top 5% research excellence during the first 3 years at Korea University
- **Granite Tower Best Teaching Award (Top 5%), Totally, 7 times**
  - Probability and Random Process KECE209-02 (2025)
  - Software Programming Basics GECT002-01 (2024), GECT002-06 (2025)
  - Computer Language and Laboratory EGRN151-07 (2019), EGRN151-06 (2021)
  - Introduction to Computers SEMI103 (2021)
  - Future Mobility Technology GEQR075 (2022)
- **Best Teaching Award (Top 20%), Totally, 7 times**
  - Probability and Random Process KECE209-02 (2021), KECE209-02 (2022)
  - Software Programming Basics GECT002-02 (2024), GECT002-08 (2024), GECT002-09 (2025)
  - Computer Language and Laboratory EGRN151-09 (2020)
  - Object-Oriented Programming SEMI104 (2021)

### Academic and University Services (International)

- **Certificate of Appreciation – IEEE Future Networks, AI/ML Working Group** 08/2025
- **Certificate of Appreciation – IEEE/IFIP WiOpt (2024)** 10/2024
- **Best Editor Award – ICT Express (Elsevier)** 07/2023
- **2022 Best Chapter Award, IEEE Vehicular Technology Society Chapter, Awarded as a Treasure** 2022
- **Appreciation Recognition – Daegu Gyeongbuk Institute of Science and Technology (DGIST)** 2021
- **Certificate of Appreciation – Department of Computer Science, University of Southern California** 2010

### Academic and University Services (Korea Regional)

- **Outstanding Contribution Award**
  - The Korean Institute of Communication Sciences and Information Sciences (KICS) 2025, 2024, 2021, 2019
  - Korean Institute of Information Scientists and Engineers (KIISE) Information Network Society 2023, 2022
  - Open Standards and ICT Association (OSIA) 2025, 2021
  - The Institute of Electronics and Information Engineers (IEIE) 2025
- **Commemorative Plaque, KICS Excellent Lecture (Title: Quantum Machine Learning and Software Technologies)** 2025
- **Best Research Group Award, KICS Research Group in Quantum Communications and Quantum Computing (Group Leader)** 2025

• **Appreciation Recognition** – Daegu Gyeongbuk Institute of Science and Technology (DGIST) 2021

### Student Awards (International)

• **IEEE TCSC Award for Excellence in Scalable Computing** (IEEE Computer Society), Awarded to *Soohyun Park* 2025  
 • **Young Scientist Award** (ICSANE 2025), Awarded to *Minji Lee* 2025  
 • **IEEE Vehicular Technology Society Student Scholarship Award (2025)**, Awarded to *Gyu Seon Kim* 2025  
 • **IEEE ICDCS Best Runner-Up Poster Paper Award (2025)**, Awarded to *Seok Bin Son, Soohyun Park* 2025

## **R&D Projects**

### University-Wide/Center Projects

• **Net-Zero CAFE Research Center** (07/2024–12/2031), **(Center Director)** ITRC (Korea Univ)  
 • **Intelligent 6G Wireless Access System Research Center** (04/2021–12/2025) 6G AI Research Center (Korea Univ)  
 • **K-Starlink: Dynamic Reconfigurable and Intelligent Space-Terrestrial Networks** (06/2021–05/2024) Basic Research Lab (Ajou)  
 • **Nano UAV Intelligence Systems Research Lab** (10/2020–08/2023) ADD Military Special Research Center (Kwangwoon)  
 • **5G/Unmanned Vehicle Research Center (5G/UV-RC)** (06/2020–12/2022) ITRC (Hanyang)  
 • **Human Resource Development for the Biomedical Unstructured Big Data Analysis** (08/2018–12/2021) ITRC (SNU-Hospital)  
 • **Novel Data Science Driven Framework for Efficient Network Design** (06/2017–05/2020) Basic Research Lab (Chung-Ang)  
 • **Intelligent Internet of Energy (IoE) Data Research Center** (02/2020–05/2020) ITRC (Kookmin)

### Industry-Funded Projects

• **Advancement Technology Development for Torpedo Deception Strategies in Submarines** (11/2022–11/2026) LIG Nex1  
 • **Advancement Tech Dev for Submarine Target Identification and Engagement Support Intelligence** (11/2022–11/2026) LIG Nex1  
 • **Mapping between Real World and VR for End-Edged Cloud Real-Time VR Servers** (09/2020–11/2025) Samsung Electronics  
 • **Research on Learning-based Swarm Mission Planning Algorithms** (03/2024–02/2025) LIG Nex1  
 • **Quantum Machine Learning-based Objection Detection for Point Cloud and its Acceleration** (12/2022–04/2024) Hyundai Motors  
 • **Routing Algorithms for LEO Satellite Networks** (12/2022–08/2023) Solvit System  
 • **Optimal Positioning Algorithms for Wide-Area Relaying Networks** (12/2022–08/2023) Solvit System  
 • **Distributed Learning Algorithms to Build AI Models with Multi-Center Clinical Data** (11/2022–02/2023) Cipherome  
 • **Cellular/Wi-Fi Handover Technology Development** (02/2022–12/2022) LG Electronics CTO Division (Smart Mobility Lab)  
 • **Research Trends in Digital Twin Applications to Autonomous Driving** (03/2022–04/2022) Hyundai NGV  
 • **Distributed Learning System Design and Implementation for Clinical Applications** (02/2022–03/2022) Cipherome  
 • **Super-Resolution Performance Optimization in Mobile Platforms** (05/2020–08/2020) Samsung SDS  
 • **Deep Learning Algorithms for mVOC Concentration Analysis** (03/2020–06/2020) Samsung Electronics (C-Lab)  
 • **Visual Recognition Software Implementation using Deep Learning Tools** (05/2019–11/2019) Hyundai Motors  
 • **A Priori Techniques Research for Efficient Multi-Edge Computing** (06/2017–12/2017) Samsung Electronics (Software Center)

### Government-Funded Projects

• **Quantum AI Empowered Second-Life Platform Technology** (07/2024–12/2031), **(Software Star-Lab)** IITP  
 • **Quantum-Empowered Spatio-Temporal Multi-Scale Digital Twin System** (03/2025–02/2028) NRF  
 • **6GARROW: 6G AI-Native Integrated RAN-Core Networks** (09/2024–08/2027) IITP  
 • **AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development** (04/2022–12/2026) IITP  
 • **Development of Integrated Development Framework that supports Automatic Neural Network Generation and Deployment optimized for Runtime Environment** (04/2021–12/2025) IITP  
 • **Quantum Hyper-Driving: Quantum-Inspired Hyper-Connected and Hyper-Sensing Autonomous Mobility** (03/2022–02/2025) NRF  
 • **Korea-Japan Joint Seminar Project for Generative and Multi-Modal AI Technologies** (10/2023–09/2024) NRF  
 • **Integrated Perception Technology Developments for Public Safety Platforms** (06/2019–05/2023) NRF  
 • **Development of Quantum Deep Reinforcement Learning Algorithm using QAOA** (10/2019–04/2022) NRF  
 • **mmWave Radar and Deep Reinforcement Learning based Optimal Policy Autonomous Driving** (06/2019–02/2022) NRF  
 • **Development of Privacy-Reinforcing Distributed Transfer-Iterative Learning Algorithm** (07/2019–12/2021) MHW  
 • **Virtual Presence in Moving Objects through 5G (PriMO-5G)** (06/2018–06/2021) IITP  
 • **Distributed Secure Platform for Scalable Clinical OMOP CDM Models** (04/2019–12/2020) MHW  
 • **mmWave High-Speed Networking Platform Design for Next-Generation Convergence Services** (06/2016–05/2019) NRF  
 • **Feasibility Study of 60 GHz IEEE 802.11ad for Virtual Reality (VR) Platforms** (04/2017–12/2017) IITP

### Government-Funded Research Institute Projects

• **Research on Trainability and Learnability in Quantum Machine Learning Models** (03.2026–10/2027) ETRI-Affiliated Institute  
 • **Discovering Quantum-Advantage Cases for Industry Adoption** (09/2025–11/2025) KTL  
 • **LEO Satellite Routing Research using Large Language Model and Reinforcement Learning** (05/2025–11/2025) ETRI  
 • **Research on Generative Quantum Machine Learning Models** (04/2025–10/2025) ETRI-Affiliated Institute  
 • **Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks** (05/2024–11/2024) ETRI  
 • **NOMA-based Resource Allocation Research in Space-Air-Ground Integrated Networks** (09/2023–11/2023) ETRI  
 • **Autonomous Intelligent COA Search Methods for Cyber-Attacks** (12/2021–11/2022) ADD  
 • **Research on Intelligent Agent-based CPS Security and Reliability** (04/2021–11/2021) TTA  
 • **Multi-GPU based Automotive HPC Platform Development** (04/2020–10/2020) ETRI  
 • **Cooperative Deep Reinforcement Learning for Online Game Multi-Agents** (04/2020–08/2020) ETRI



- *Verification Testbed Implementation for Privacy-Preserving Trust Data Generation* (10/2019–11/2019) ETRI
- *Measurement and Analysis of Multi-Task GPU Scheduling Delays* (05/2019–10/2019) ETRI
- *Probabilistic Decision Making and Econometric Methods for Micro-Grid* (05/2017–04/2019) KEPCO Research Institute
- *GPU Scheduling Performance Analysis under Queueing Delay Considerations* (05/2018–10/2018) ETRI
- *Improving Massive Deep Learning Training via Computation and Communication Acceleration* (04/2018–10/2018) ETRI
- *Parsing Techniques for Artificial Neural Network (ANN) Data Processing* (09/2017–11/2017) ETRI

### University of Southern California, Selected Projects

- *Video Aware Wireless Networks (VAWN) Research Program* Intel Labs, Verizon Wireless, Cisco Systems
- *60 GHz Real-Time Wireless Video Broadcasting* Disney Research Zürich
- *Annenberg Graduate Fellowship Award* University of Southern California

### Korea University (KU), Selected Projects

- *Towards Agentic AI-Enabled Edge IoT Systems for Proactive Physical World Sensing and Control* (08/2025–07/2026) PKU-KU Joint Research Grant (Collaboration with Prof. Kaigui Bian at Peking Univ)
- *AI Teaching Assistant Research for LLM-based Large-Scale Education* (10/2024–02/2025) KU University College
- *Autonomous Mobility Control using Quantum Deep Learning* (03/2023–02/2024) KU Insung Research Grant
- *Mobile Access Algorithm Design using Economic Theory and AI* (03/2020–02/2021) KU Future Research Grant

## Selected Publications

- **11667+ Citations** (H-index: 52+, i10-index: 250+), obtained from Google Scholar Profile (as of April 27, 2026)
- **Journals and Magazines:** **168** publications; among them, **124** IEEE/Nature/APS publications

### Books

- *Fundamentals of 6G Communications and Networking*, Springer (2024) (Editors: X. Lin, J. Zhang, Y. Liu, J. Kim)

### Selected Papers

#### ■ Magazines

- [CM.accepted] Sustainable 6G Non-Terrestrial Networks: Enabling Net-Zero Operation via Renewable Energy Integration and Intelligent Control, *IEEE Communications Magazine*, v(n):ppp-ppp (H. Lee, S. Park, S. Jung, J. Kim)
- [CM.accepted] Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective, *IEEE Communications Magazine*, v(n):ppp-ppp (G. S. Kim, J. Chung, S. Park, S. Jung, T. Q. Duong, J. Kim)
- [NM.accepted] Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks, *IEEE Network*, v(n):ppp-ppp (H. Lee, G. S. Kim, E. J. Roh, S. Jung, S. Park, T. Q. Duong, J. Kim)
- [MM'25.10-12] Quantum Jump to Virtual Worlds: High-Quality Multiple Virtual Meta-Space Realization in Metaverse, *IEEE MultiMedia*, 32(4):37–44 (S. Park, J. Kim)
- [CIM'25.08] Quantum-Eyes: Scalable Quantum Convolutional Neural Networks for Low-Overhead Object Detection, *IEEE Computational Intelligence Magazine*, 20(3):63–74 (J. Kim, E. J. Roh, C. Im, S. Park)
- [CM'24.12] The Matrix: Quantum AI for Interacting Two Worlds in Prioritized Metaverse Spaces, *IEEE Communications Magazine*, 62(12):97–103 (S. Park, H. Baek, J. Kim)
- [CM'24.10] Quantum Multi-Agent Reinforcement Learning is All You Need: Coordinated Global Access in Integrated TN/NTN Cube-Satellite Networks, *IEEE Communications Magazine*, 62(10):86–92 (S. Park, G. S. Kim, Z. Han, J. Kim)
- [CM'24.06] Quantum Multi-Agent Reinforcement Learning for Autonomous Mobility Cooperation, *IEEE Communications Magazine*, 62(6):106–112 (S. Park, J. P. Kim, C. Park, S. Jung, J. Kim)
- [IC'23.09-10] EQuaTE: Efficient Quantum Train Engine for Run-Time Dynamic Analysis and Visual Feedback in Autonomous Driving, *IEEE Internet Computing*, 27(5):24–31 (S. Park, H. Feng, C. Park, Y. K. Lee, S. Jung, J. Kim)
- [CSM'22.06] Recent and Future Evolution of Wi-Fi, *IEEE Communications Standards Magazine*, 6(2):8–11 (E. Au, L. Wilhelmsson, T. Baykas, J. Kim)
- [PIEEE'21.05] Communication-Efficient and Distributed Learning Over Wireless Networks: Principles and Applications, *Proceedings of the IEEE*, 109(5):796–819 (J. Park, S. Samarakoon, A. Elgabri, J. Kim, M. Bennis, S.-L. Kim, M. Debbah)
- [CM'19.03] New Challenges of Wireless Power Transfer and Secured Billing for Internet of Electric Vehicles, *IEEE Communications Magazine*, 57(3):118–124 (L. Park, S. Jeong, D. S. Lakew, J. Kim, S. Cho)
- [VTM'17.03] The Useful Impact of Carrier Aggregation: A Measurement Study in South Korea for Commercial LTE-Advanced Networks, *IEEE Vehicular Technology Magazine*, 12(1):55–62 (S. Lee, S. Hyeon, J. Kim, H. Roh, W. Lee)

#### ■ Top-Tier Journals/Conferences in Networks and Systems

- [TMC.accepted] Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient Cube-Satellite Networks, *IEEE Transactions on Mobile Computing*, 25(n):ppp-ppp (S. Park, G. S. Kim, Y. Cho, S. Jung, Z. Han, J. Kim)
- [TAES'26.03] Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket, *IEEE Transactions on Aerospace and Electronic Systems*, 62:3838–3863 (G. S. Kim, J. Chung, S. Jung, S. Park, J. Kim)
- [TAES.'26.01] Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles, *IEEE Transactions on Aerospace and Electronic Systems*, 62:682–703 (G. S. Kim, S. Park, S. Jung, D. Mohaisen, J. Kim)
- [TMC'26.01] Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks, *IEEE Transactions on Mobile Computing*, 25(1):1200–1218 (G. S. Kim, Y. Cho, J. Chung, S. Park, S. Jung, Z. Han, J. Kim)

- [IPDPS'25] AQUA: Hardware-Agnostic Qubit Allocation for Quantum Multi-Programming, *IEEE Int'l Parallel and Distributed Processing Symposium* (X. Piao, J. Shim, J. Kim, J. Kim) (Acceptance Rate: 24.71%)
- [WiOpt'25] Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach, *IEEE/IFIP Int'l Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (G. S. Kim, J. Chung, T. Q. Duong, S. Park, J. Kim)
- [NOMS'25] Joint Multi-Agent Reinforcement Learning and Message-Passing for Distributed Multi-UAV Network Management using Conflict Graphs, *IEEE/IFIP Network Operations and Management Symposium* (Y. Cho, H. Lee, S. Park, J. Kim)
- [TON'25.08] Slimmable Federated Reinforcement Learning for Energy-Efficient Proactive Caching, *IEEE Transactions on Networking*, 33(4):2079–2094 (H. Baek, G. S. Kim, S. Park, A. F. Molisch, J. Kim)
- [TAES'25.08] Quantum Multiagent Reinforcement Learning for Joint Cube-Satellites and High-Altitude Long-Endurance Aerial Vehicles in SAGIN, *IEEE Transactions on Aerospace and Electronic Systems*, 61(4):9490–9510 (G. S. Kim, Y. Cho, S. Park, S. Jung, J. Kim)
- [TMC'25.02] Fast Quantum Convolutional Neural Networks for Low-Complexity Object Detection in Autonomous Driving Applications, *IEEE Transactions on Mobile Computing*, 24(2):1031–1042 (E. J. Roh, H. Baek, D. Kim, J. Kim)
- [WiOpt'24] Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management, *IEEE/IFIP Int'l Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (S. Lee, G. S. Kim, S. Park, J. Kim)
- [TON'24.12] Spatio-Temporal Multi-Metaverse Dynamic Streaming for Hybrid Quantum-Classical Systems, *IEEE/ACM Transactions on Networking*, 32(6):5279–5294 (S. Park, H. Baek, J. Kim)
- [TMC'24.12] Joint Quantum Reinforcement Learning and Stabilized Control for Spatio-Temporal Coordination in Metaverse, *IEEE Transactions on Mobile Computing*, 23(12):12410–12427 (S. Park, J. Chung, C. Park, S. Jung, M. Choi, S. Cho, J. Kim)
- [TWC'24.03] Joint User Clustering, Beamforming, and Power Allocation for mmWave-NOMA with Imperfect SIC, *IEEE Transactions on Wireless Communications*, 23(3):2025–2038 (B. Lim, W. Yun, J. Kim, Y.-C. Ko)
- [TMC'24.01] Learning Location from Shared Elevation Profiles in Fitness Apps: A Privacy Perspective, *IEEE Transactions on Mobile Computing*, 23(1):581–596 (U. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen)
- [TON'23.12] SlimFL: Federated Learning with Superposition Coding over Slimmable Neural Networks, *IEEE/ACM Transactions on Networking*, 31(6):2499–2514 (W. Yun, Y. Kwak, H. Baek, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim)
- [WiOpt'22] Cooperative Video Quality Adaptation for Delay-Sensitive Dynamic Streaming using Adaptive Super-Resolution, *IEEE/IFIP Int'l Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (M. Choi, W. Yun, J. Kim)
- [INFOCOM'22] Joint Superposition Coding and Training for Federated Learning over Multi-Width Neural Networks, *IEEE INFOCOM* (H. Baek, W. Yun, Y. Kwak, S. Jung, M. Ji, M. Bennis, J. Park, J. Kim) (Acceptance Rate: 19.93%)
- [TMC'22.05] Supremo: Cloud-Assisted Low-Latency Super-Resolution in Mobile Devices, *IEEE Transactions on Mobile Computing*, 21(5):1847–1860 (J. Yi, S. Kim, J. Kim, S. Choi)
- [TMC'21.06] A Personalized Preference Learning Framework for Caching in Mobile Networks, *IEEE Transactions on Mobile Computing*, 20(6):2124–2139 (A. Malik, K. S. Kim, J. Kim, W.-Y. Shin)
- [TWC'21.04] Probabilistic Caching and Dynamic Delivery Policies for Categorized Contents and Consecutive User Demands, *IEEE Transactions on Wireless Communications*, 20(4):2685–2699 (M. Choi, A. F. Molisch, D.-J. Han, D. Kim, J. Kim, J. Moon)
- [ICDCS'20] Understanding the Potential Risks of Sharing Elevation Information on Fitness Applications, *IEEE Int'l Conference on Distributed Computing Systems* (Ü. Meteriz, N. F. Yildiran, J. Kim, D. Mohaisen) (Acceptance Rate: 17.98%)
- [TWC'20.12] Joint Distributed Link Scheduling and Power Allocation for Content Delivery in Wireless Caching Networks, *IEEE Transactions on Wireless Communications*, 19(12):7810–7824 (M. Choi, A. F. Molisch, J. Kim) (IEEE ComSoc MMTC Best Journal Paper Award)
- [TWC'19.12] Markov Decision Policies for Dynamic Video Delivery in Wireless Caching Networks, *IEEE Transactions on Wireless Communications*, 18(12):5705–5718 (M. Choi, A. No, M. Ji, J. Kim)
- [TWC'19.10] Dynamic Power Allocation and User Scheduling for Power-Efficient and Delay-Constrained Multiple Access Networks, *IEEE Transactions on Wireless Communications*, 18(10):4846–4858 (M. Choi, J. Kim, J. Moon)
- [TMC'19.07] Seamless Dynamic Adaptive Streaming in LTE/Wi-Fi Integrated Network under Smartphone Resource Constraints, *IEEE Transactions on Mobile Computing*, 18(7):1647–1660 (J. Koo, J. Yi, J. Kim, M. A. Hoque, S. Choi)
- [ICDCS'18] ShmCaffe: A Distributed Deep Learning Platform with Shared Memory Buffer for HPC Architecture, *IEEE Int'l Conference on Distributed Computing Systems* (S. Ahn, J. Kim, E. Lim, W. Choi, A. Mohaisen, S. Kang) (Acceptance Rate: 20.63%)
- [JSAC'18.11] SGCO: Stabilized Green Crosshaul Orchestration for Dense IoT Offloading Services, *IEEE Journal on Selected Areas in Communications*, 36(11):2538–2548 (N.-N. Dao, D.-N. Vu, W. Na, J. Kim, S. Cho)
- [JSAC'18.06] Wireless Video Caching and Dynamic Streaming under Differentiated Quality Requirements, *IEEE Journal on Selected Areas in Communications*, 36(6):1245–1257 (M. Choi, J. Kim, J. Moon)
- [ACM-MM'17] REQUEST: Seamless Dynamic Adaptive Streaming over HTTP for Multi-Homed Smartphone under Resource Constraints, *ACM Int'l Conference on Multimedia* (J. Koo, J. Yi, J. Kim, M. A. Hoque, S. Choi) (Acceptance Rate: 27.63%)
- [TON'16.08] Quality-Aware Streaming and Scheduling for Device-to-Device Video Delivery, *IEEE/ACM Transactions on Networking*, 24(4):2319–2331 (J. Kim, G. Caire, A. F. Molisch)
- [MobiSys'10] Energy-Efficient Rate-Adaptive GPS-based Positioning for Smartphones, *ACM Int'l Conference on Mobile Systems, Applications, and Services* (J. Paek, J. Kim, R. Govindan) (Acceptance Rate: 19.84%)

■ Top-Tier Journals/Conferences in AI (especially, Quantum AI, Reinforcement Learning, and LLM)

- [KDD'26] (Double Blind Review), *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (H. Ahn, E. J. Roh, S. Park, W. Saad, H.-C. Lee, J. Kim)
- [ACL'26] Let LLM Tutors Ask First: Proactive LLM-Based Tutoring at Scale in a 1,500-Student Online Classroom, *Annual Meeting*



- [ICASSP'26] How Can Quantum Deep Learning Improve Large Language Models?, *IEEE Int'l Conference on Acoustics, Speech, and Signal Processing* (E. J. Roh, H. Ahn, S. Y.-C. Chen, S. Park, J. Kim)
- [CommPhy'26.00] Fourier Analysis Perspective on Quantum Neural Networks, *Communications Physics (Nature)*, v(n):ppp-ppp (S. Oh, E. J. Roh, A. Vasilakos, S. Park, A. Vasilakos, S. Park, J. Kim)
- [PRA'26.02] Understanding Qubit Trajectories in Parametrized Quantum Circuits, *Phy. Rev. A*, 113:022609 (S. Oh, C. Im, S. Park, J. Kim)
- [CIKM'25] Quantum-Amplitude Embedded Adaptation for Parameter-Efficient Fine-Tuning in Large Language Models, *ACM Conference on Information and Knowledge Management* (E. J. Roh, J. Kim) **(Acceptance Rate: 30.63%)**
- [CIKM'25] Filtered One-Shot Training for Quantum Architecture Search, *ACM Conference on Information and Knowledge Management* (S. B. Son, S. Y.-C. Chen, J. Kim, S. Park) **(Acceptance Rate: 30.63%)**
- [CIKM'25] LLM-based Interactive Coding Education via Predictive Query Management and Student-Centered Fine-Tuning: Design and Implementation with 1500-Student Class Data, *ACM Conference on Information and Knowledge Management* (G. Youn, J. Lee, J. Kim, C. Yoo) **(Acceptance Rate: 30.63%)**
- [ICASSP'25] Quantum Reinforcement Learning for Coordinated Satellite Systems, *IEEE Int'l Conference on Acoustics, Speech, and Signal Processing* (G. S. Kim, S. Chen, S. Park, J. Kim)
- [CIKM'24] Hands-On Introduction to Quantum Machine Learning, *ACM Conference on Information and Knowledge Management* (S. Y.-C. Chen, J. Kim), **Tutorial**
- [TNNLS'24.02] Hierarchical Deep Reinforcement Learning-based Propofol Infusion Assistant Framework in Anesthesia, *IEEE Transactions on Neural Networks and Learning Systems*, 35(2):2510–2521 (W. Yun, M. Shin, D. Mohaisen, K. Lee, J. Kim)
- [CIKM'23] Quantum Split Learning for Privacy-Preserving Information Management, *ACM Conference on Information and Knowledge Management* (S. Park, H. Baek, J. Kim) **(Acceptance Rate: 27.44%)**
- [CIKM'23] Logarithmic Dimension Reduction for Quantum Neural Networks, *ACM Conference on Information and Knowledge Management* (H. Baek, S. Park, J. Kim) **(Acceptance Rate: 27.44%)**
- [AAAI'23] Quantum Multi-Agent Meta Reinforcement Learning, *AAAI* (W. Yun, J. Park, J. Kim) **(Acceptance Rate: 19.61%)**
- [CIKM'22] Hierarchical Reinforcement Learning using Gaussian Random Trajectory Generation in Autonomous Furniture Assembly, *ACM Conference on Information and Knowledge Management* (W. Yun, D. Mohaisen, S. Jung, J.-K. Kim, J. Kim) **(Acceptance Rate: 29.64%)**
- [IJCAI'19] Randomized Adversarial Imitation Learning for Autonomous Driving, *Int'l Joint Conference on Artificial Intelligence* (M. Shin, J. Kim) **(Acceptance Rate: 17.89%)**

#### ■ Journals

- [EAAI.accepted] A Comprehensive Review on Quantum Deep Neural Networks for Prognostics and Health Management, *Engineering Applications of Artificial Intelligence*, v:ppp (M. S. Jha, S. Y.-C. Chen, C. Kulkarni, J. Kim)
- [TVT.accepted] Time-to-Collision Aware Autonomous Driving Motion Control under Risky Scenario Considerations, *IEEE Transactions on Vehicular Technology*, v(n):ppp-ppp (S. Kim, J. Kim, J. Kim, S. Jung)
- [FGCS'26.09] GraMA: A Gradient Matrix-Guided Assignment Method for Solving Qubit Mapping Problems, *Future Generation Computer Systems*, 182:108485 (X. Piao, J. Kim, J.-K. Kim)
- [IOTJ'26.06] Quantum Federated Gradient Aggregation using the Parameter-Shift Rule, *IEEE Internet of Things Journal*, 13(11):ppp-ppp (J. Chung, C. Im, S. Park, J. Kim, W. Lee)
- [TVT'26.04] Large-Scale Battery-Conscious Collision-Free Path-Finding for Sustainable and Autonomous Multi-AGV Mobility Control, *IEEE Transactions on Vehicular Technology*, 75(4):5364–5375 (Y. Cho, S. Ahn, S. Park, J. Kim)
- [IOTJ'26.03] Gateway-Assisted Neural Angular Routing for Hierarchical Satellite Networks, *IEEE Internet of Things Journal*, 13(6):12322–12339 (J. Jang, I.-S. Cho, M. Shin, J. Kim, S. Jung)
- [IOTJ'26.02] Hybrid Large Language Model and Reinforcement Learning for Energy-Efficient Multisatellite Scheduling: Boosting the Performance from Scratch, *IEEE Internet of Things Journal*, 13(3):5379–5392 (H. Ahn, G. S. Kim, I.-S. Cho, S. Jung, J. Kim)
- [AC'26.02] Collaborative Learning Architecture for Autonomous Excavator Planning and Execution, *Automation in Construction*, 182:106742 (J. Cho, M. Shin, J. Kim, S. Jung) **(2024 JCR Ranked 1st, in the area of Civil Engineering (1/184))**
- [TIV'25.11] Adaptive Quantum Federated Learning for Autonomous Surveillance Multi-Drone Networks, *IEEE Transactions on Intelligent Vehicles*, 10(11):5055–5059 (S. Park, C. Park, S. Jung, J. Kim)
- [JS'25.10] Joint Scalable Quantum Convolutional Neural Network and Reverse-Fidelity Training for High-Accurate Recognition in Unmanned Aerial Vehicle Surveillance, *The Journal of Supercomputing*, 81(16):1495 (E. J. Roh, J. Kim, S. Jung, S. Park)
- [TASE'25.09] Adaptive Excavation Automation in Complex Soil Environments using Reinforcement Learning, *IEEE Transactions on Automation Science and Engineering*, 22:21181–21194 (M. Shin, J. Cho, J. Kim, S. Jung)
- [ESWA'25.09] Correlation-Assisted Spatio-Temporal Reinforcement Learning for Stock Revenue Maximization, *Expert Systems with Applications*, 289:128361 (J. Chung, M. Kim, S. Min, H. Choi, S. Park, J. Kim)
- [IOTJ'25.08] Joint Interference Approximation and Guard-Band Management for Spectrum-Efficient Integrated NTN-TN Networks, *IEEE Internet of Things Journal*, 12(15):32220–32236 (J. Jang, J. Kim, J. Kim, S. Jung)
- [IOTJ'25.08] Carbon-Aware Edge Computing for Internet of Everything Networks: A Digital Twin Approach, *IEEE Internet of Things Journal*, 12(15):29240–29251 (D. V. Huynh, S. R. Khosravirad, V. Sharma, J. Kim, B. Canberk, T. Q. Duong)
- [JCN'25.08] Stabilized Classification Control using Multi-Stage Quantum Convolutional Neural Networks for Autonomous Driving, *Journal of Communications and Networks*, 27(4):264–272 (E. J. Roh, S. Park, S. Jung, J. Kim)
- [JCN'25.08] Special Issue on Quantum Technologies for Communication Systems, *Journal of Communications and Networks*, 27(4):217–221 (S. Y. Shin, Z. Han, S. Ali, S. Y.-C. Chen, Y. Liu, S. Park, J. Kim)
- [IOTJ'25.07] Quantum Reinforcement Learning for Lightweight LEO Satellite Routing, *IEEE Internet of Things Journal*, 12(14):28986–

- [IOTJ'25.07] Privacy-Preserving Uncertainty Calibration using Perceptual Encryption in Cloud-Edge Collaborative Artificial Intelligence of Things, *IEEE Internet of Things Journal*, 12(13):25424–25441 (I. Ahmad, J. Kim, S. Shin)
- [IOTJ'25.07] Joint Quantum Reinforcement Learning and Neural Myerson Auction for High-Quality Digital-Twin Services in Multitier Networks, *IEEE Internet of Things Journal*, 12(13):23722–23735 (S. Park, G. S. Kim, J. Kim)
- [NN'25.07] Quantum Federated Learning with Pole-Angle Quantum Local Training and Trainable Measurement, *Neural Networks*, 187:107301 (S. Park, H. Lee, S. B. Son, S. Jung, J. Kim)
- [JS'25.07] Quantum Infidelity Codistillation for Fast and Accurate Distributed Quantum Machine Learning, *The Journal of Supercomputing*, 81(10):1151 (S. Oh, J. Kim, J. Park, H. Baek, H. Lee, J. Kim, S.-L. Kim)
- [IOTJ'25.06] Entanglement-Controlled Quantum Federated Learning, *IEEE Internet of Things Journal*, 12(11):18318–18330 (S. Park, H. Lee, S. Jung, J. Park, M. Bennis, J. Kim)
- [JS'25.06] SQUAD: Software Testing for Quantum Distributed Learning Software, *The Journal of Supercomputing*, 81(9):1071 (S. Park, J. H. Cho, H. J. Yook, G. S. Jhun, Y. K. Lee, J. Kim, S. Park)
- [TVT'25.05] Dynamic Quantum Federated Learning for UAV-based Autonomous Surveillance, *IEEE Transactions on Vehicular Technology*, 74(5):8158–8170 (S. Park, S. B. Son, S. Jung, J. Kim)
- [TNSM'25.04] Intelligent Extra Resource Allocation for Cooperative Awareness Message Broadcasting in Cellular-V2X Networks, *IEEE Transactions on Network and Service Management*, 22(2):1677–1689 (S. Jung, J.-H. Kim, J. Kim)
- [JS'25.02] Hybrid Quantum-Classical 3D Object Detection using Multi-Channel Quantum Convolutional Neural Network, *The Journal of Supercomputing*, 81(3):455 (E. J. Roh, J. Y. Shim, J. Kim, S. Park)
- [EL'25.01] Fast Batch Gradient Descent in Quantum Neural Networks, *IET Electronics Letters*, 61(1):e70162 (J. Y. Shim, J. Kim)
- [IOTJ'24.12] Markov Decision Policies for Distributed Angular Routing in LEO Mobile Satellite Constellation Networks, *IEEE Internet of Things Journal*, 11(23):38744–38754 (S. Park, G. S. Kim, S. Jung, J. Kim)
- [TIV'24.11] Neural Myerson Auction for Truthful and Distributed Mobile Charging in UAV-Assisted Digital-Twin Networks, *IEEE Transactions on Intelligent Vehicles*, 9(11):7326–7338 (S. Jung, H. Baek, J. Kim)
- [Access'24.10] Sensing-to-Sky Intermittent Connectivity Realization for LTE-Enabled Drone Platforms: Embedded Design, Measurement Study, and Positioning Applications, *IEEE Access*, 12:137360–137372 (J. Kim, S. Park, U. Jo, T. Kim, S. Jung, J. Kim)
- [FGCS'24.10] AQUA: Analytics-driven Quantum Neural Network (QNN) User Assistance for Software Validation, *Future Generation Computer Systems*, 159:545–556 (S. Park, H. Baek, J. W. Yoon, Y. K. Lee, J. Kim)
- [ETRI'24.10] Trends in Quantum Reinforcement Learning: State-of-the-Arts and the Road Ahead, *ETRI Journal*, 46(5):748–758 (S. Park, J. Kim) **(Top Cited Article, among work published in 2024)**
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- [Access'24.04] Quantum Reinforcement Learning for Spatio-Temporal Prioritization in Metaverse, *IEEE Access*, 12:54732–54744 (S. Park, H. Baek, J. Kim)
- [TGCN'24.03] Joint Delay-Sensitive and Power-Efficient Quality Control of Dynamic Video Streaming using Adaptive Super-Resolution, *IEEE Transactions on Green Communications and Networking*, 8(1):103–117 (M. Choi, W. Yun, S. Son, S. Park, J. Kim)
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- [ETRI'23.10] Two Tales of Platoon Intelligence for Autonomous Mobility Control: Enabling Deep Learning Recipes, *ETRI Journal*, 45(5):735–745 (S. Park, H. Lee, C. Park, S. Jung, M. Choi, J. Kim)
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- [OJCS'23.09] Real-Time High-Quality Visualization for Volumetric Contents Rendering: A Lyapunov Optimization Framework, *IEEE Open Journal of the Computer Society*, 4:243–252 (H. Baek, R. Lee, S. Jung, J. Kim, S. Park)
- [TIV'23.08] Multi-Agent Reinforcement Learning for Cooperative Air Transportation Services in City-Wide Autonomous Urban Air Mobility, *IEEE Transactions on Intelligent Vehicles*, 8(8):4016–4030 (C. Park, G. S. Kim, S. Park, S. Jung, J. Kim)
- [NN'23.08] Stereoscopic Scalable Quantum Convolutional Neural Networks, *Neural Networks*, 165:860–867 (H. Baek, W. Yun, S. Park, J. Kim)
- [IOTJ'23.06] Quantum Multiagent Actor-Critic Neural Networks for Internet-Connected Multirobot Coordination in Smart Factory



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- [ComNet'23.04] Truthful and Performance-Optimal Computation Outsourcing for Aerial Surveillance Platforms via Learning-based Auction, *Computer Networks*, 225:109651 (S. Jung, J.-H. Kim, D. Mohaisen, J. Kim)
- [CIBM'23.04] Deep Reinforcement Learning-based Propofol Infusion with a 3,000-subject Dataset in Anesthesia, *Computers in Biology and Medicine*, 156:106739 (W. Yun, M. Shin, S. Jung, J. Ko, H.-C. Lee, J. Kim)
- [Access'23.03] Audio-to-Visual Cross-Modal Generation of Birds, *IEEE Access*, 11:27719–27729 (J. Y. Shim, J. Kim, J.-K. Kim)
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- [TII'22.10] Cooperative Multi-Agent Deep Reinforcement Learning for Reliable Surveillance via Autonomous Multi-UAV Control, *IEEE Transactions on Industrial Informatics*, 18(10):7086–7096 (W. Yun, S. Park, J. Kim, M. Shin, S. Jung, D. Mohaisen, J.-H. Kim)
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- [ISJ'22.06] Securing Heterogeneous IoT with Intelligent DDoS Attack Behavior Learning, *IEEE Systems Journal*, 16(2):1974–1983 (N.-N. Dao, T. Phan, U. Sa'ad, J. Kim, T. Bauschert, D.-T. Do, S. Cho)
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- [Nature'22.01] Feasibility Study of Multi-Site Split Learning for Privacy-Preserving Medical Systems under Data Imbalance Constraints in COVID-19, X-Ray, and Cholesterol Dataset, *Nature Scientific Reports*, 12:1534 (Y. J. Ha, G. Lee, M. Yoo, S. Jung, S. Yoo, J. Kim)
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- [ISJ'21.09] Intelligent Active Queue Management for Stabilized QoS Guarantees in 5G Mobile Networks, *IEEE Systems Journal*, 15(3):4293–4302 (S. Jung, J. Kim, J.-H. Kim)
- [Access'21.09] Spatio-Temporal Split Learning for Privacy-Preserving Medical Platforms: Case Studies with COVID-19 CT, X-Ray, and Cholesterol Data, *IEEE Access*, 9:121046–121059 (Y. J. Ha, M. Yoo, G. Lee, S. Jung, S. Choi, J. Kim, S. Yoo)
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- [JCN'21.04] Stabilized Adaptive Sampling Control for Reliable Real-Time Learning-based Surveillance Systems, *Journal of Communications and Networks*, 23(2):129–137 (D. Kim, S. Park, J. Kim, J. Y. Bang, S. Jung)
- [JCN'21.04] Dynamic Video Delivery using Deep Reinforcement Learning for Device-to-Device Underlaid Cache-Enabled Internet-of-Vehicle Networks, *Journal of Communications and Networks*, 23(2):117–128 (M. Choi, M. Shin, J. Kim)
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## Academic Activities and Research Supervision

### Research Supervision

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- Dr. Soohyun Park (03/2019–08/2023 (PhD), 09/2023–02/2024 (Postdoc)), **Sookmyung Women's University (Professor)**
- Dr. Hankyul Baek (03/2021–02/2024 (PhD), 03/2024–03/2025 (Postdoc)), ETRI AI Safety Institute (Researcher)
- Dr. Hyunsoo Lee (03/2021–02/2026 (PhD)), LG Electronics (Researcher)
- Dr. Gyu Seon Kim (03/2023–02/2026 (PhD), 03/2026–04/2026 (Postdoc)), RealtimeVisual (Researcher, *Military Service Exception*)
- Dr. Seok Bin Son (03/2022–02/2027 (Expected) (PhD))
- Dr. Emily Jimin Roh (03/2024–02/2027 (Expected) (PhD))

#### ■ M.S. Alumni

- Kyeongseon Kim (09/2017–08/2019), POSTECH (Ph.D. Student in Electrical Engineering)
- Dohyun Kwon (03/2018–02/2020), Hyundai Marine & Fire Insurance (Researcher)
- Dohyun Kim (03/2018–02/2020), Naver Webtoon (Researcher)
- MyungJae Shin (03/2018–02/2020), Naver (Researcher)
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- Jaehyun Chung (09/2023–08/2025), Korea University (Ph.D. Student in Electrical and Computer Engineering)
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- Dr. Minseok Choi (09/2018–02/2019, jointly with Prof. Andreas F. Molisch (USC)), **Kyung Hee University (Professor)**
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### IEEE Activities (Membership, Editorial Boards, and Services)

- Senior Member (2018–), Member (2006–2017)
- Associate Editor (2025–), **ACM Computing Surveys**
- Associate Editor (2025–), **IEEE Communications Surveys and Tutorials**
- Editor (2023–), **IEEE Internet of Things Journal**
- Associate Editor (2020–), **IEEE Transactions on Vehicular Technology**
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